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Started	Tuesday, 13 October 2026, 8:00 PM
Completed	Tuesday, 13 October 2026, 8:07 PM
Duration	6 mins 42 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search requires the list to be sorted.
b. Linear search divides the list into halves.
c. Linear search is more efficient than binary search on large lists.
d. Linear search can be applied to both sorted and unsorted lists. ✓

Question **2**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search can find multiple instances of the target element
b. Binary search does not require dividing the list
c. Binary search has a lower time complexity on large, sorted lists ✓
d. Binary search works on unsorted lists

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Binary search algorithm is not efficient when the data elements more than 1500 ✓
b. Must use a sorted array
c. Requirement of sorted array is expensive when a lot of insertion and deletions are needed
d. There must be a mechanism to access middle element directly

Question **4**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Sequential search
b. Linear search
c. Binary search ✓
d. Hash search

Question **5**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search continues in the left sublist
b. The search ends successfully ✓
c. The search continues in the right sublist
d. The list is sorted

Question **6**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Binary search
b. Hash search
c. Linear search ✓
d. Both (1) & (3)

Question **7**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Jump search
b. Hashing
c. Binary search
d. Linear search ✓

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

- a. Merge search
- b. Bubble search
- c. Quick search
- d. Linear search ✓

Question **9**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

- a. O(1)
- b. O🥐
- c. O(n log n)
- d. O(log n) ✓

Question **10**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

- a. In the left sublist
- b. In the right sublist ✓
- c. At the beginning of the list
- d. In the middle of the list

Question **11**

Incorrect

Mark 0.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

- a. Linear search
- b. Interpolation search
- c. Binary search
- d. Hash search ✗

Question **12**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

- a. By comparing each element sequentially
- b. By using a hash function
- c. By dividing the list length by two ✓
- d. By starting from the first element

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

- a. When the list is very small
- b. When the list is unsorted
- c. When the list is sorted ✓
- d. When the list contains duplicate elements

Question **14**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

- a. 500
- b. 10 ✓
- c. 100
- d. 1000

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

- a. It requires the list to be sorted ✓
- b. It is slower than linear search for small lists
- c. It does not work with negative numbers
- d. It can only be applied to large lists